



Shader Building (Snow - Unreal Engine 5)

The Arctic Epoch - Singapore (Group Project)

Project Description (Group Project)

In 2035, global climate change gets more and more acute. Extreme weather conditions happen first on a regional scale in many different parts of the world but human society does not take it seriously. Eventually, a global-scale snowstorm comes and the Earth enters a new ice age. The snowstorm freezes up the world within a month. Famine happens, riots follow, and in consequence, human society is destroyed.

This project captures the scenery in Singapore a few months after the society collapses.

Concept

An icy post-apocalyptic city, Singapore

Used Software

Unreal Engine 5, Houdini

Time

14 Weeks

Goal

To create realistic cities covered by snow with snow & ice shaders.

Role

Shader Technical, Lighting Artist
I mainly focused on a realistic snow generator building with Blueprint Script, and an icicle generator with Houdini.
Also, video editing with SFX(license free) was done.

Team

Shader Technical Artist, Lighting Artist - Christine Jung
Technical Artist, Look-Dev Artist - Robin Zheng
Technical Environment Artist - Caiyu Zhang
Environment Artist, Technical Artist - Yuxuan Wu
Technical Artist, Look-Dev Artist - Laura Yang

Piled Snow Generator - UE5 Blueprint Script



Snow Generator

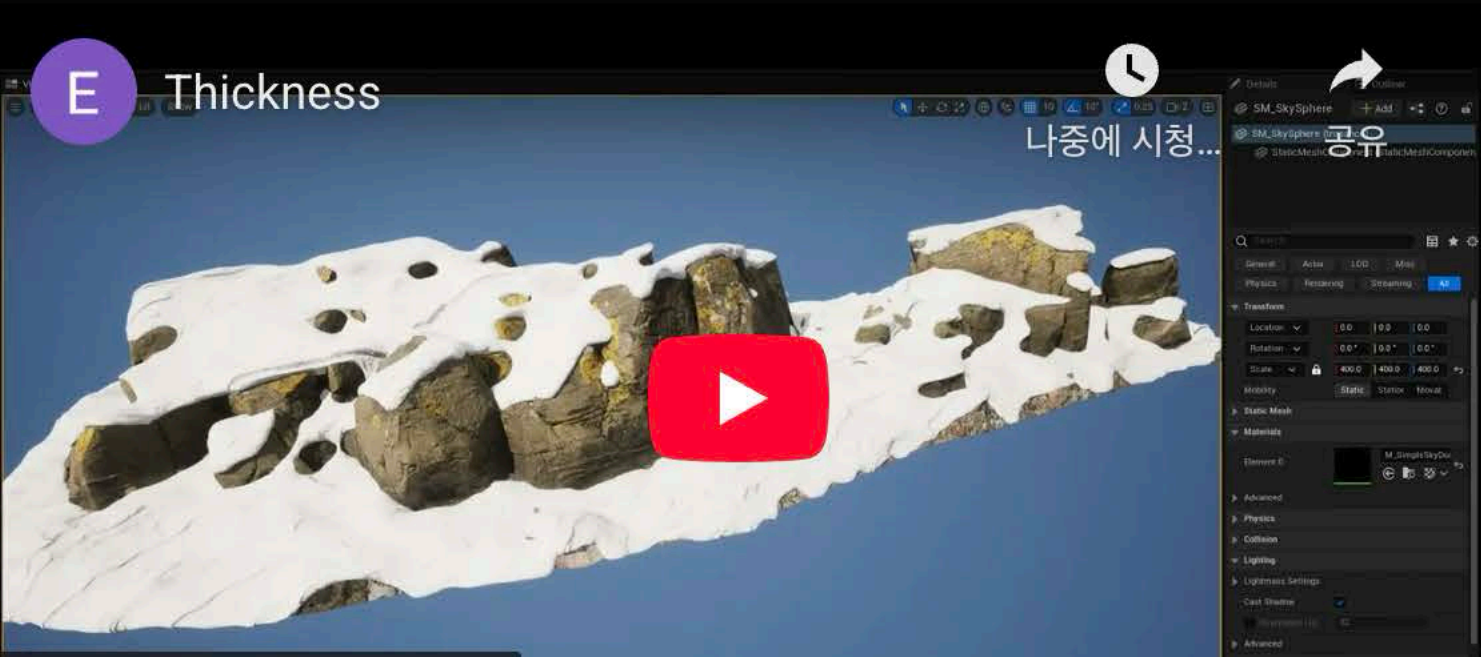
Requirement

- 1. To create realistic piled snow with a thickness
- 2. Has to be generated according to every object
- 3. In snow storm environment, snow has to be swepted

- Used Unreal Engine 5 Blueprint Script to create the snow generator

- Available features:

Snow coverage / Snow sparse rate / Snow (sphere) size / Snow



Texture / Displacement scale & Strength / Mesh cache, and so on

다음에서 보기:  YouTube

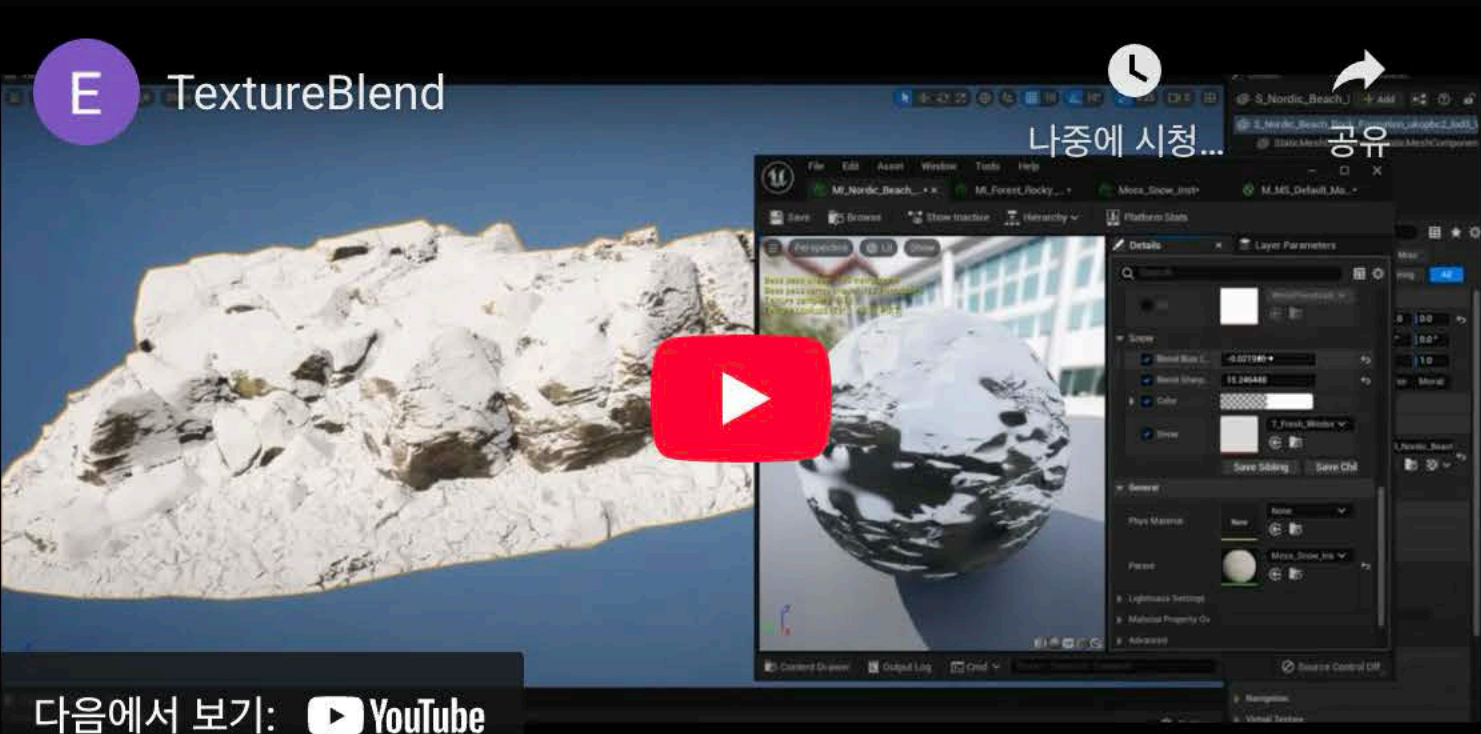
Problem

- 1. Too heavy to apply the generator to big scaled models or high poly models
- 2. Has to cache the mesh and generate the snow every time whenever the transform and scale values are different



Solution

- 1. Generate snow on very small-sized objects(0.02) for the efficiency
- 2. Bake the output as a static mesh



Snow Texture Blend Material

Requirement

- 1. To create snow texture that blends into objects' texture to have a more realistic look

- Used Unreal Engine 5 Blueprint to create it
- Created a material that can be applied to every object's parent material
- Available features:
 - Blend bias(snow amount) / Bled sharpness(snow sharpness)



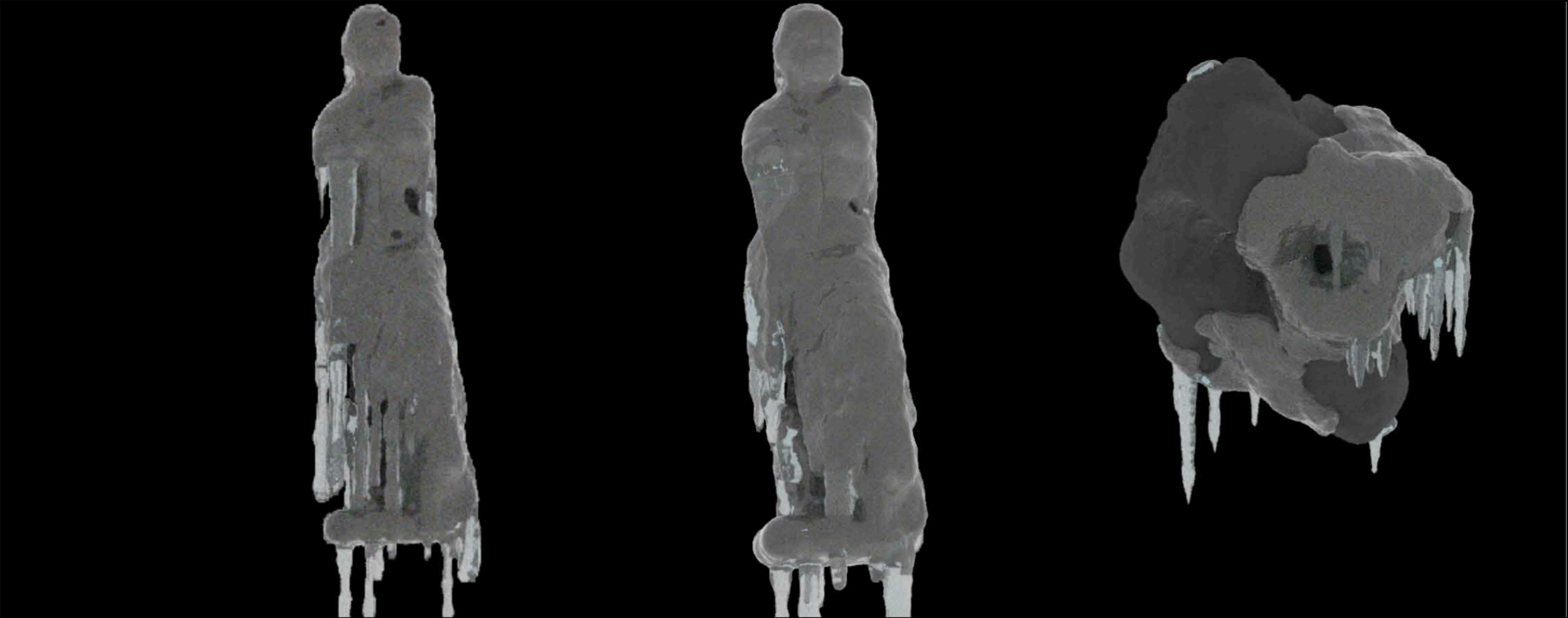
Icicle Generator

Created an ice & icicle generator using Houdini. Although we only used the final frame only as the output, but the icicle grows(simulates) corresponding to the original mesh.

You can adjust the parmeters of :

- Birth rate (# of seeds for spawning)
- Growth speed (length)
- Thickness, shape
- Ice coverage





Lighting

Lighted to acheive the look of disastrous atmosphere but with a bit of hope





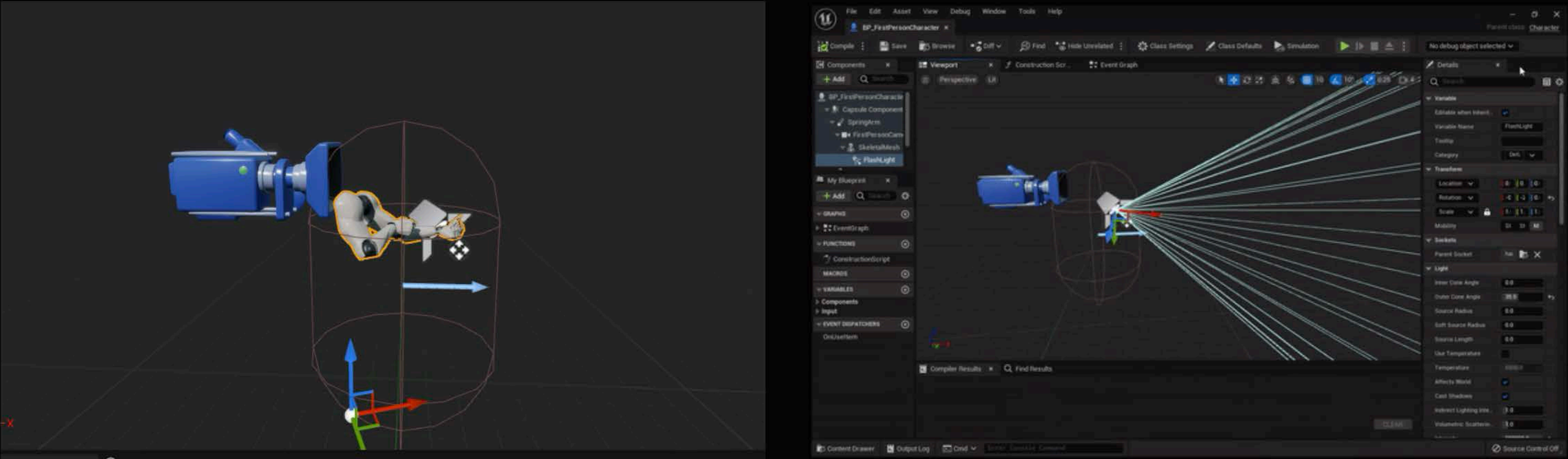
Rim light for the main asset, the Merlion

Flashlight

Created a flashlight effect using Blueprint, so that the light follows the rotation of the camera, and shake corresponding to the breathing & walking animation of the player character.



walking animation of the player character.



Ice shader - Problem Solving



Ice shader - Problem Solving



Problem 1
This ice shader was created by one of the teammates, but a problem occurred when the ice was interacting with the snowstorm. The shader used the background, or the scene that were positioned behind ice object, to look translucent. It was working fine without the snow storm, but it looked like it's fully transparent when the snow storm(sprites of 2D images) was behind the ice.

Problem 2
With the same reason, flickering happened with the ice object in the distance. For example, Merlion statue looks like it was flickering.

Solution
Instead of transparent shader, I chose to make to shader opaque. Yet, I was able to achieve volumetric ice look by offsetting UV coordinate with camera vector.





Snowflake Generator

Requirement

- 1. To create realistic snowflake to use for the intro scene
- 2. Have parameters to tweak to find the right snowflake shape

- Referred [this paper](#)(A local cellular model for snow crystal growth - Clifford A. Reiter *) to get the knowledge of how the snowflakes grow and how to simulate it.
- Available features:
 - 1. Alpha(weighting Factor): center seed cell diffusion weight
 - 2. Beta(background leve): distance from the initial cell
 - 3. Gamma(constant): adds constant to the growth of the ice (cells surrounding seed cell)

*Because the team decided to switch the deliverable from cinematic to playable demo, we did not use this snowflake for the intro scene.





$\alpha = 1.0, \beta = 0.6, \gamma = 0.0001$



$\alpha = 2.6, \beta = 0.4, \gamma = 0.0001$



$\alpha = 2.5, \beta = 0.4, \gamma = 0.0001$



$\alpha = 2.003, \beta = 0.4, \gamma = 0.0001$



$\alpha = 1.0, \beta = 0.4, \gamma = 0.0001$